

Final Project Report – EK 210 A7 Group C

Executive Summary

This report provides details about a project involving the engineering of a room occupancy monitor. This device reads and displays the number of people in a room at a given time, and triggers an alert system when the occupancy crosses the predetermined threshold. The room occupancy is measured using two sets of IR emitters and receivers placed across from each other in two separate enclosures on either side of the doorway. When people walk through the monitor, they block the emission from reaching the receiver. A computer program calculates the occupancy given that the direction at which a person enters the doorway indicates if the room count increases or decreases. The alert system consists of a buzzer and a LED to account for accessibility needs; people who are hard of hearing or seeing can still get the most out of the device. During the testing process, it was found that the accuracy of the monitor was at least 96% in 50 trials for different walking speeds. Furthermore, the alert system was effective in that the buzzer was audible from at least 20 feet while the LED was visible from at least 10 feet.

Introduction

It has always been crucial to uphold room occupancy regulations in commercial settings. During fires, for example, evacuations are more efficient when rooms are not overcrowded. Additionally, pandemic situations require room occupancy regulations to enable social distancing and limit disease spread. The monitor presented in this report allows people to view the room occupancy so individuals can take steps to ensure their safety during emergencies.

Problem Statement

Design a device that monitors the room population and alerts people when it exceeds a threshold so that security personnel can uphold maximum occupancy regulations.

Key Objectives/Metrics

The first objective of this project was to maintain and display the number of people in a room. The metric for this was that the count should have at least 90% accuracy. The second objective was that the device should alert people when the maximum occupancy was crossed, via a buzzer and an LED. The metric for this was that the buzzer should be audible at 20 feet and a red LED should be visible from 10 feet. The third objective was that there should be some manual control for security personnel. The metric was that the power switch and sensor

alignment detector should have 100% accuracy. Refer to Appendix A for a full list of objectives, metrics, and constraints, along with a pairwise comparison chart that orders the objectives by importance.

Important/Innovative Design Elements and Basis for Selection

During set up, the device indicates to users if the emitter/receiver sensors are aligned, allowing for easy set up and debugging of technical issues. The device's alarm system incorporates both a buzzer and a red warning LED. This combination provides both visual and audio indicators for when occupancy is surpassed, increasing accessibility of the alarm system. An LCD screen is used to display the current occupancy of the room to people about to enter, clearly informing people on whether they should enter. Refer to Appendix B for the glass box analysis and a morph chart for this product with detailed information about the functions and corresponding means. Additionally, Appendix C contains a complete list of parts used in the final product along with their prices.

Description of Final Product

The final product consists of two separate boxes, one containing the IR emitters and the other containing the receiver, Arduino, LCD screen, buzzer, and red LED. Both of the boxes have command strips attached to them for easy mounting. To set up the device, users place command strips on the walls by the doorway and will attach the boxes' strips to the walls. To use the device, the user uses the switch on each box to supply power. If the IR emitter and receiver are not aligned with each other, the LCD displays a "NOT ALIGNED" message indicating to users that they need to adjust their sensor orientation. Once the system is powered on, the device displays a count of the occupancy inside the room. The count changes when a person walks in between the two boxes, blocking the emission. The emitter/receiver pairs are parallel with each other; the count increases/decreases depending on which emitter/receiver pair is blocked first (indicating if a person is entering or exiting). If the count exceeds the maximum occupancy limit, a buzzer/alarm is triggered and a red warning LED turns on. A "LIMIT EXCEEDED" message is also displayed on the LCD along with the count. Refer to Appendix D for CAD drawings and final product images; Appendix E contains an electronics sketch; also, Appendix F contains a code flow chart which explains how the program converts the sensory information into count.

The device also has some manual control features for security personnel. To reset the count to 0, users can turn off the box with the LCD and turn it back on. To manually adjust the

count to a specific value, users can move their hands across the IR sensors to change the count. Lastly, the battery life for the device is 6 hours. Appendix G contains the mathematical calculation for the power budget.

Evaluation of Results

As shown in Appendix H, 50 trials were conducted to test the accuracy of the occupancy count at different walking/running speeds. 25 trials tested entering and the other 25 tested leaving. Slower walking at 2 to 3 mph resulted in 100% (50/50) accuracy of count updates. Fast walking at 4 to 5 mph resulted in 98% (49/50) accuracy. Running at 6 to 7 mph resulted in 96% accuracy (48/50). Audibility of the buzzer and visibility of the red LED and LCD screen were tested using 4 participants. All participants were able to clearly hear the buzzer at the 20 ft minimum. All participants were able to clearly see the red LED and screen at the 10 ft minimum. The accuracy of the manual controls (power switch and alignment detector) were also tested using 25 trials. All 25 trials passed resulting in 100% accuracy.

Lessons Learned

A key lesson we learned when 3D printing our CAD design was to expect margins of error with the 3D print and the CAD dimensions. Although a CAD model could show a hole perfectly fitted for an LED, once printed, the dimensions of said hole could be off, leading it to be too small for the LED. We also learned that, when designing a CAD model it is important to consider how each part will be held in place (screws, glue, etc.). The orientation of the device must also be accounted for. Lastly, the device's code should account for all possible edge cases. When testing the device, consider all possible scenarios a user might encounter.

References

- Datasheet for IR Break Beam Sensor – 5mm: <https://www.adafruit.com/product/2168>
- Datasheet for Piezo Buzzer - PS1240: <https://www.adafruit.com/product/160>
- Datasheet for Arduino Uno R3: <https://www.adafruit.com/product/4806>
- Datasheet for Standard LCD 16x2:
https://www.adafruit.com/product/181?gad_source=1&gclid=CjwKCAiAmsurBhBvEiwA6e-WPD0umg8Rp3OmjcqUqJfoAUd2YIhgslmNspZuZi0MzYW_VZo_d-MUZBoCsWwQAvD_BwE
- Datasheet for LED – Basic Red 3mm: <https://www.sparkfun.com/products/533>

Appendix

1) Appendix A: Objectives/Metrics/Constraints and Pairwise Comparison Chart

OBJECTIVES

Objective 1: Maintain/store count of people and display

Metric: Occupancy count should have at least 90% accuracy.

Objective 2: Alert users when count crosses maximum.

Metric: Buzzer must be audible at a minimum of 20 ft. Red LED and screen must be visible at a minimum of 10 ft.

Objective 3: Manual control for personnel to turn on/off and align the two boxes.

Metric: 100% consistency for power switch and alignment detector.

CONSTRAINTS

Constraint 1: Device should use invisible light to detect changes in occupancy rather than visible light or mechanical means.

Metric: IR sensors must be used rather than visible light sensors or mechanical means.

Constraint 2: Device should be of a reasonable size to not be an obstruction to the room.

Metric: Device's dimensions must be within 4 x 4 x 4 inches.

Constraint 3: Device should have its own power source that can reliably run its functions.

Metric: Device must use its own battery and have a battery life of at least 6 hours.

Table 1: This table depicts the pairwise comparison chart which ranks all the objectives

	Store Count	Alert	Inform	Power	Size	Invisible Light	Total
Store Count	—	1	1	1	1	1	5
Alert	0	—	1	1	1	1	4
Inform	0	0	—	1	1	1	3
Power	0	0	0	—	0	0	0
Size	0	0	0	1	—	0	1
Invisible Light	0	0	0	1	1	—	2

2) Appendix B: Glass Box and Morph Chart

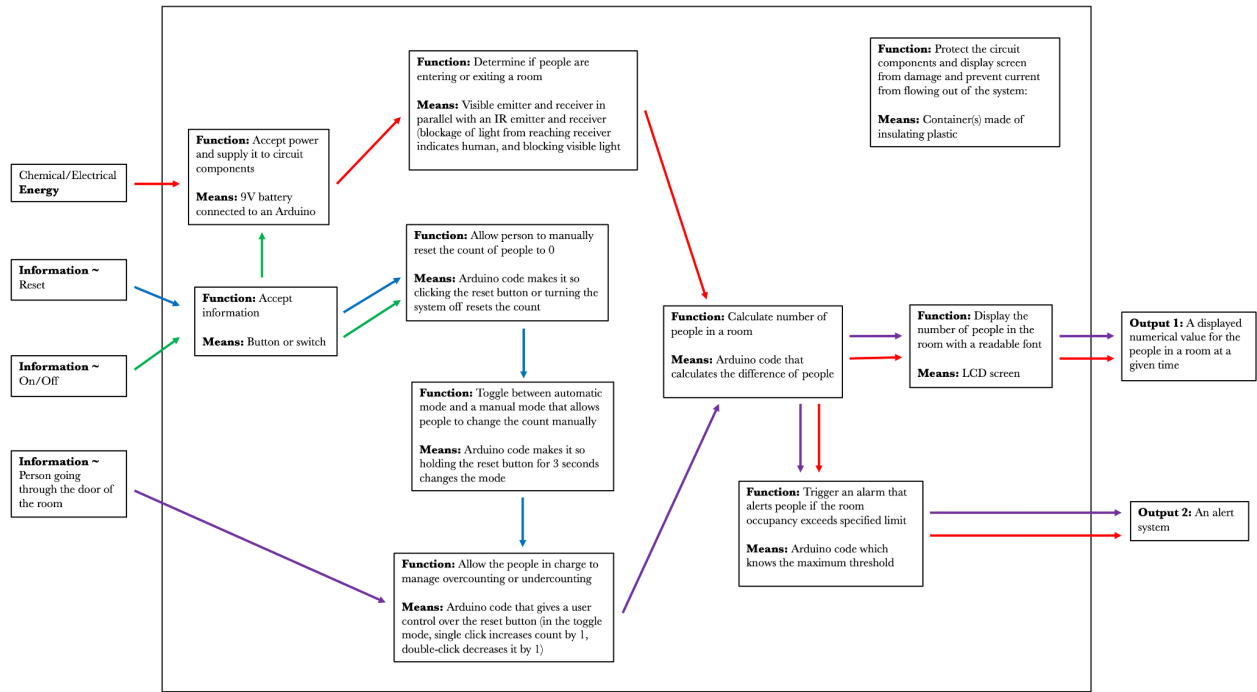


Figure 1: This image depicts the glass box analysis for the monitor. It defines key functions along with their means.

Table 2: The table below reflects some of the important functions of the monitor along with the means considered to accomplish them.

Principal Functions	Means				
	1	2	3	4	5
Supply/accept power	Rechargeable battery with cable	Linking to AC electric circuit in the wall via wires	9-volt batteries		
Sense a signal that represents the number of people entering and exiting a room	Audio-based sensors (measure sound of footsteps)	IR sensors (detect infrared radiation emitted by people)	Camera (visual detection of people)	Light sensors (people might block/reflect light when they enter)	Pressure sensor (detect when people sit down on chairs in rooms)
Convert the signal into useful information/calculate the number of people in a room	Formula programmed into Arduino (difference between number of people entering and exiting)	Machine learning model (uses sensor data and trends to predict number of people)	Fusion of sensors (using multiple to improve the accuracy of the formula/ML)		
Display the number of people in the room on a screen that is easy to visualize	LCD screen (displays calculated number on the device on the wall)	Mobile/web application (allows users to access the room population remotely)	LEDs (different colors of light correspond to different room occupancies)	SMS alert (notify people in the building/room when occupancy is maximum)	
Sound an alarm if the number of people exceed the maximum occupancy	Buzzer connected to Arduino (turns on when max occupancy is exceeded)	Mobile/web application alert system (vibration motors in phones are activated)	LEDs (red color starts flashing to indicate dangerous situation)		
Hold components together for safety	Plastic insulating container (limits electricity flow outside circuit)	Wall mount (keeps the device away from physical damage from ground)	Plastic bag (can seal loose parts while acting as an insulator)	Rubber bands (tightly keep loose parts together while insulating)	

3) Appendix C: Parts

Table 3: This data table shows all the parts used in the final design along with their costs.

Part Name	Quantity	Cost Per Unit	Total Cost
IR Break Beam Sensors	2	5.95	11.90
Slide Switches	2	0.76	1.58
PKCell 9V Battery	2	2.00	4.00
Piezo Buzzer	1	1.50	3.00
Arduino Uno	1	22.50	22.50
LCD	1	9.95	9.95
LED Red	1	0.45	0.45
Enclosure	2	3.38	6.76
Wires	25	.10	2.50
Total Sum			62.64

4) Appendix D: CAD Drawings

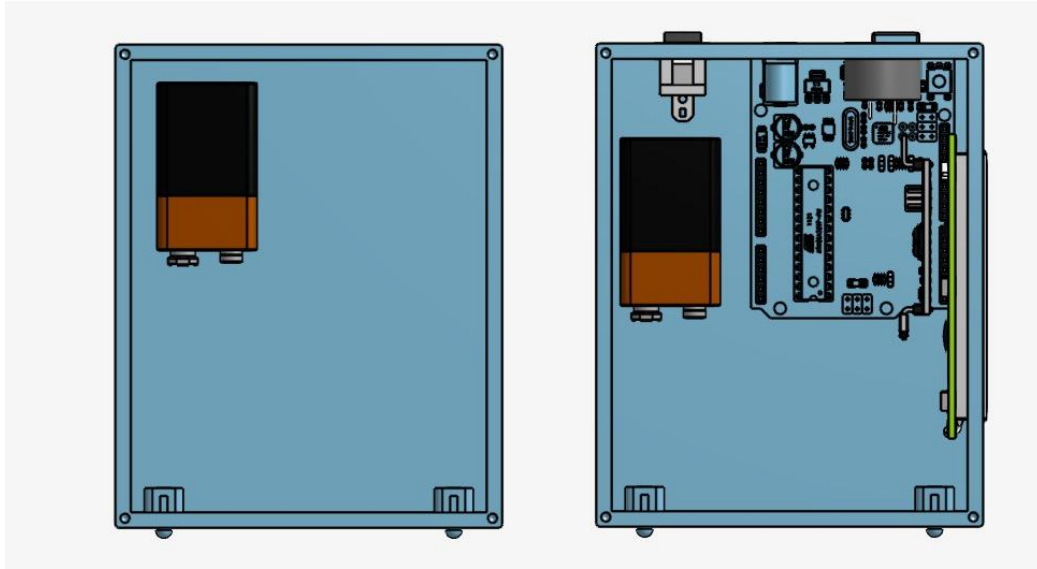


Figure 2: The box on the left contains the emitters and a battery to power them, while the box on the right contains the receivers, Arduino, LCD, and alert system.

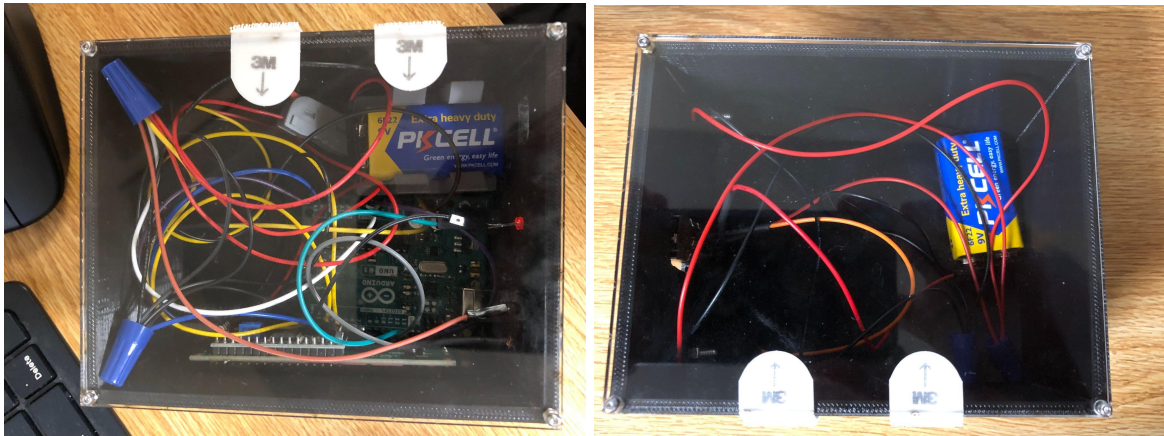


Figure 3: The box on the left depicts the interior of the final IR receiver box, and the box on the right depicts the interior of the final IR emitter box (battery holder not pictured). Note that the enclosures are 3D printed plastic while the lids are made of acrylic.

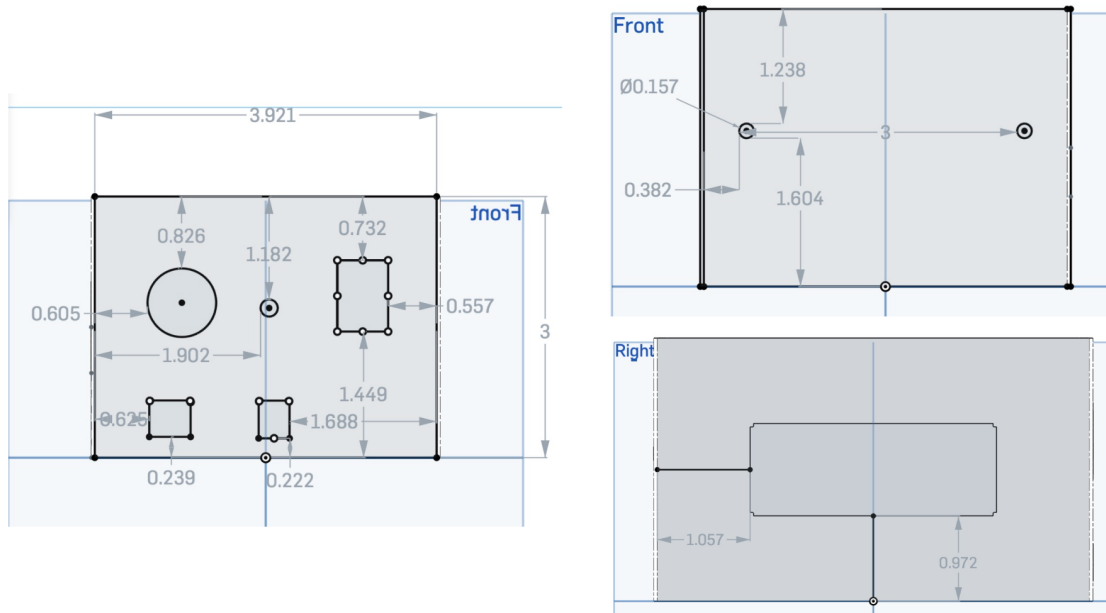


Figure 4: This image depicts the dimensions of the enclosure and the parts inside it. Note that the dimensions are specific to the box with the receivers, though both boxes are the same size.

5) Appendix E: Electronics Sketch

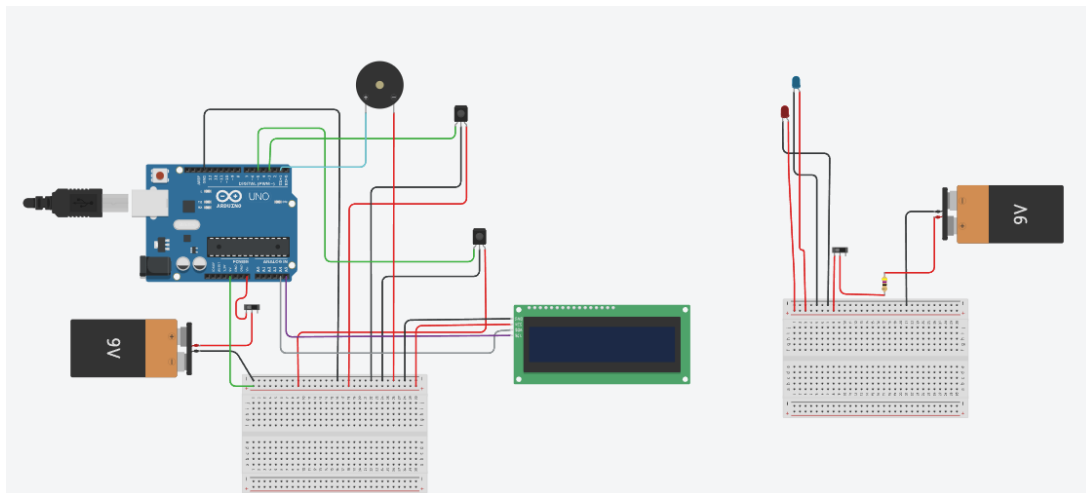


Figure 5: The sketch on the left represents the wiring in the box with the IR receivers, alert system, and LCD. The sketch on the right represents the wiring in the box with the emitters.

6) Appendix F: Code Flow Chart

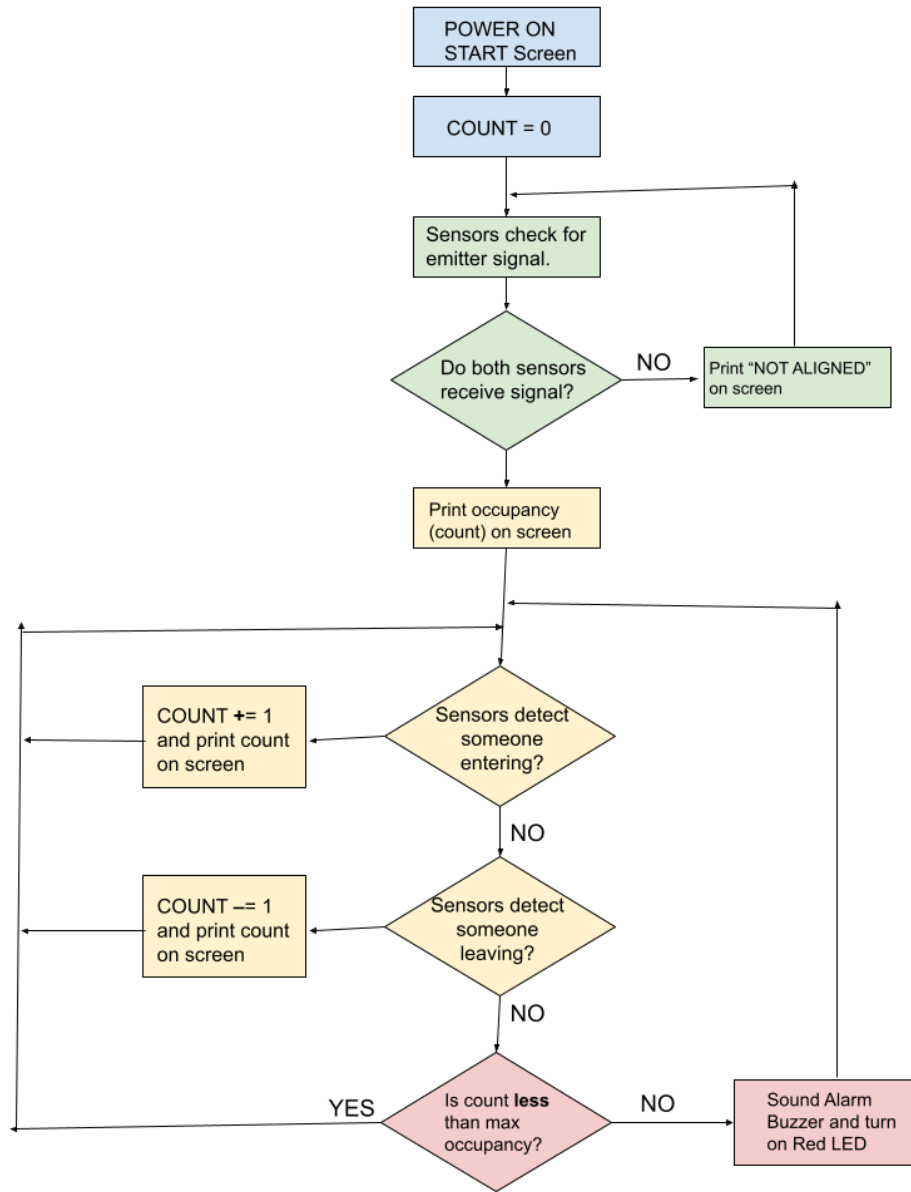


Figure 6: The sketch above is a visual representation of the device's code structure. Rhombus blocks represent conditionals. Rectangular blocks represent commands. The green blocks are responsible for the sensors' initial alignment detection. The yellow is for the mechanism which updates the count/occupancy. The red is for the alarm system when the count exceeds the maximum capacity/limit. Users can power off the device anytime, stopping the code execution.

7) Appendix G: Power Budget Calculation

Table 4: This data table shows the voltage across, current through, and power delivery of the room occupancy monitor's key circuit components.

Part Name	Quantity	Voltage(V)	Current(mA)	Power(mW)
IR Break Beam Sensors	2	5V	70 mA	350 mW
LCD	1	5V	20 mA	100mW
LED Red	1	5V	17 mA	85 mW
IR Break Beam Emitters	2	2.42V	67.8 mA	164.076mW
Piezo Buzzer	1	5V	20 mA	100mW
Total				799 mW

8) Appendix H: Results

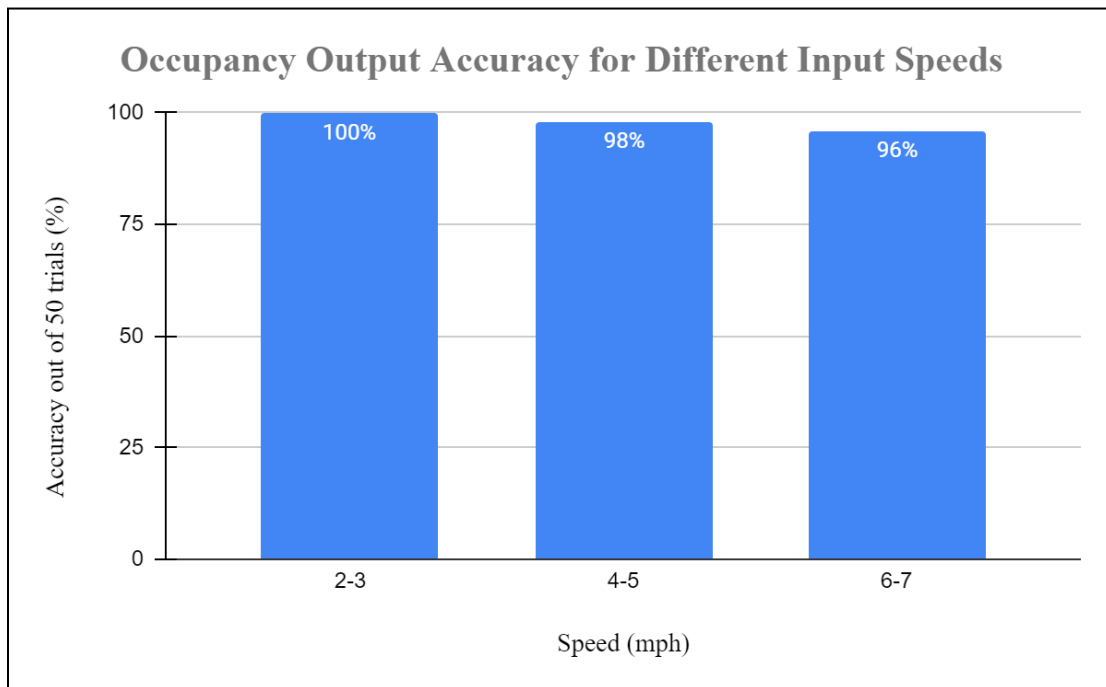


Figure 7: This graph depicts the accuracy of the room occupancy measured by the device for someone walking (2-3 mph), jogging (4-5 mph), and running (6-7 mph) through the device.

Table 5: This data table shows the audibility of the buzzer used in the alert at up to 25 feet

Audibility of Buzzer as Distance Increases				
	Audible?			
Distance from Device (ft)	Person 1	Person 2	Person 3	Person 4
0	Yes	Yes	Yes	Yes
5	Yes	Yes	Yes	Yes
10	Yes	Yes	Yes	Yes
15	Yes	Yes	Yes	Yes
20	Yes	Yes	Yes	Yes
25	Yes	No	Yes	Yes

Table 6: This data table shows the visibility of the LED used in the alert at up to 15 feet

Visibility of red warning LED and LCD Screen as Distance increases				
	Clearly Visible?			
Distance from Device (ft)	Person 1	Person 2	Person 3	Person 4
0	Yes	Yes	Yes	Yes
5	Yes	Yes	Yes	Yes
10	Yes	Yes	Yes	Yes
15	Yes	Yes	Yes	Yes

Table 7: This data table shows the accuracy of the switch and alignment detector used for manual control out of 25 trials

Manual User Controls Accuracy (out of 25 trials)	
	Accuracy (%)
Power Switch	100%
Alignment Detection on LCD Screen	100%